E-Commerce Customer Intelligence Case Study

Customer Behaviour and Revenue Optimization Analysis

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Tools Used	Microsoft Excel, Power BI
Dataset	Online Retail — UCI Machine Learning Repository
Dataset Size	541,910 transactions

2. Business Problem Statement

This analysis aims to answer the following business questions —

1. Which customers generate the most revenue?
2. Which countries drive the highest sales?
3. What time of year generates maximum revenue?
4. Which products sell the most?
5. Which customers are at risk of leaving and should be targeted for retention?

3. Data Cleaning Summary

The original dataset contained 541,910 rows and 8 columns — Invoice, StockCode, Description, Quantity, InvoiceDate, Price, CustomerID and Country.

The following cleaning steps were performed —

6. Negative Quantities — 10,624 rows removed as they represented product returns not actual sales.
7. Blank CustomerIDs — 133,361 rows removed as customers without IDs cannot be tracked for RFM analysis.
8. Non-numeric StockCodes — 1,555 rows removed as they represented fees and admin entries like POST, BANK CHARGES and DOT — not real products.

Final clean dataset — 396,370 rows ready for analysis.

Note: Revenue was calculated as a derived column using the formula Revenue = Quantity x Price. This single column powered all revenue analyses throughout the project.

4. Key Insights

4.1 Revenue Overview

The business generated total revenue of £8.76 million from 396,370 transactions after data cleaning.

4.2 Seasonal Revenue Trends

I kept both year and month grouping to avoid combining data across different years which would distort seasonal trend analysis. The dataset begins from December 2010, so 2010 data is incomplete and should not be compared directly with 2011.

Revenue peaks during September, October and November driven by festive season demand and winter shopping. February and April show lowest revenue indicating post-festive slowdown. This suggests the business should increase inventory and marketing spend before September.

Despite December being the festive month, November records higher revenue because online shoppers order gifts in advance to ensure timely delivery before Christmas. This is a unique characteristic of e-commerce businesses compared to physical retail stores.

4.3 Revenue by Country

United Kingdom generates 25x more revenue than Netherlands. The business has extreme geographic concentration with UK accounting for over 80% of revenue. This creates a critical business risk — if the UK market faces any downturn, the entire business would be severely impacted. The company should urgently diversify into other European markets.

Top international markets after UK —

- Netherlands — £283,889
- EIRE — £257,296
- Germany — £205,569

4.4 Top Products

Paper Craft Little Birdie is the top selling product by revenue at £168,469. However, this product also recorded the largest return of 80,995 units suggesting potential quality or supply chain issues.

Top selling products are dominated by home decor and gifting items suggesting the business primarily serves the gifting and home decoration market. This aligns with the seasonal revenue peaks observed during festive months.

4.5 Customer Analysis

The top 10 customers out of 8,671 total customers contribute 17.35% of total revenue. This indicates a moderately concentrated customer base where a small group of high value customers drive a significant portion of business. This strongly suggests the need for a VIP customer retention program.

A notable example is Customer 12346 — a One Time Big Spender with the following RFM metrics —

- Recency — 326 days since last purchase (Score: 1)
- Frequency — 1 order only (Score: 1)
- Monetary — £77,183 total spend (Score: 3)

They spent a huge amount but only once and never came back. This is called a ONE TIME BIG SPENDER — very common in B2B businesses where companies place one large bulk order.

4.6 RFM Customer Segmentation

RFM Scoring Framework used in this analysis —

- Score 8-9 — VIP Customer
- Score 6-7 — Loyal Customer
- Score 4-5 — Potential Customer
- Score 2-3 — At Risk Customer
- Score 1 — Lost Customer

VIP customers represent only 29.3% of the customer base but contribute 87.2% of total revenue. Losing even a small number of VIP customers would devastate business revenue. An immediate VIP retention program with exclusive offers, early access, and dedicated support is critical for business survival.

631 at risk customers should be targeted with personalised re-engagement campaigns such as discount codes, loyalty points, or personalised product recommendations. Re-engaging existing customers is significantly more cost effective than acquiring new ones.

5. Business Recommendations

9. VIP Customer Retention Program — VIP customers generate 87% of revenue. The business should immediately launch exclusive loyalty programs, early access offers and dedicated support for these customers. Losing even a few VIP customers would severely impact revenue.

10. Seasonal Inventory Planning — Revenue peaks in September to November. The business should increase inventory and marketing spend before September to capitalize on festive season demand.

11. Re-engage At Risk Customers — 631 at risk customers should be targeted with personalised discount codes and product recommendations. Re-engaging existing customers is 5x more cost effective than acquiring new ones.

12. International Market Expansion — While UK dominates revenue, Netherlands and EIRE show strong potential. Targeted marketing campaigns in these countries could significantly grow international revenue and reduce dangerous geographic concentration.

13. Product Focus — Home decor and gifting products dominate sales. The business should expand this product category and increase stock before the festive season.

14. Quality Check on Top Products — Paper Craft Little Birdie recorded the largest product return of 80,995 units. A quality and supply chain review is recommended for this product.

6. Dashboard

The Power BI dashboard below was built to visualize all key findings from this analysis.

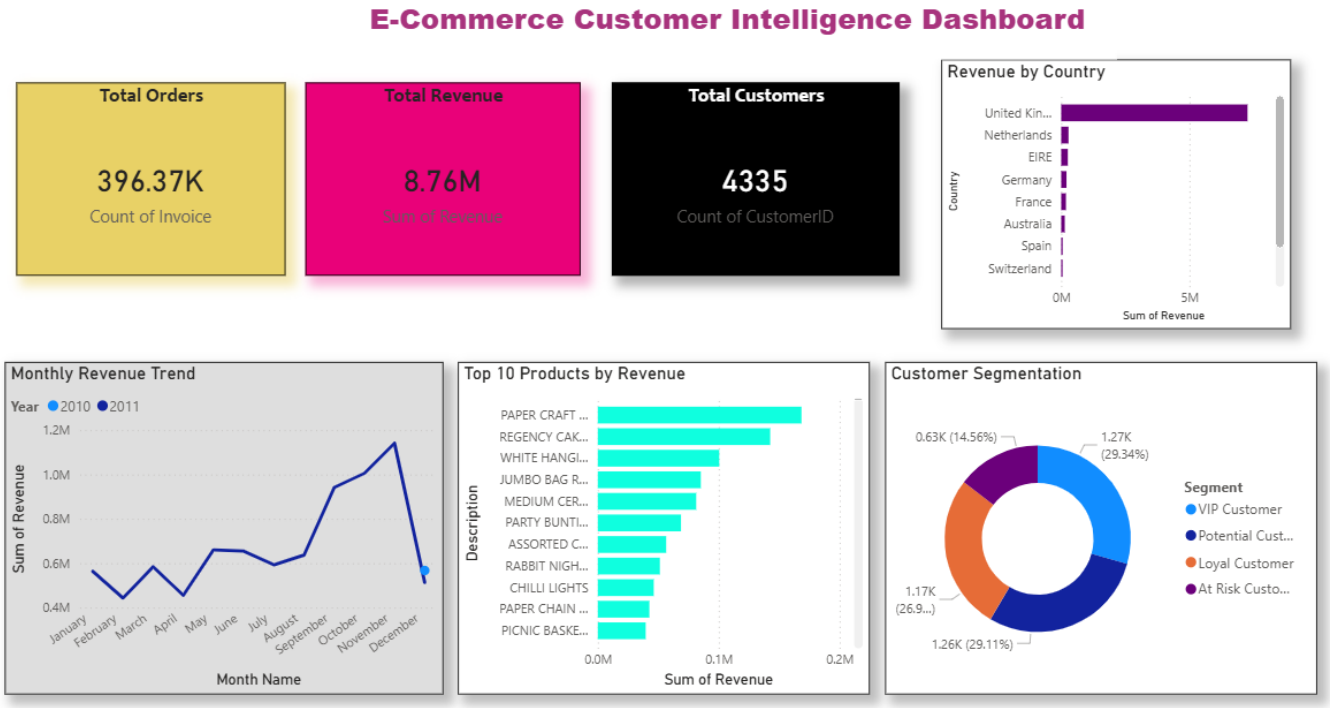

Figure 1 — E-Commerce Customer Intelligence Dashboard built using Microsoft Power BI showing revenue trends, country analysis, top products and customer segmentation.